

NSU CSE499 Final Project

GTA V Roleplay Server

Qbox Framework Implementation

Complete Documentation Package

Generated: December 09, 2025

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1. Main Presentation

NSU CSE499 Final Project Presentation

GTA V Roleplay Server - Qbox Framework

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1. Introduction

Project Title

NSU CSE499 - GTA V Roleplay Server

Team Members

[Add your team member names here]

Supervisor

[Add supervisor name here]

Project Duration

[Add project timeline here]

2. Project Overview

What is FiveM?

- Multiplayer modification framework for Grand Theft Auto V
- Allows custom servers with unique gameplay modes
- Supports custom scripts, resources, and modifications

Project Objectives

- Develop a fully functional roleplay server for GTA V
- Implement realistic job systems and economy
- Create immersive player interactions
- Build a scalable, modular server architecture

Target Audience

- Roleplay gaming community
- GTA V multiplayer enthusiasts
- Players seeking realistic life simulation gameplay

3. Technologies & Frameworks

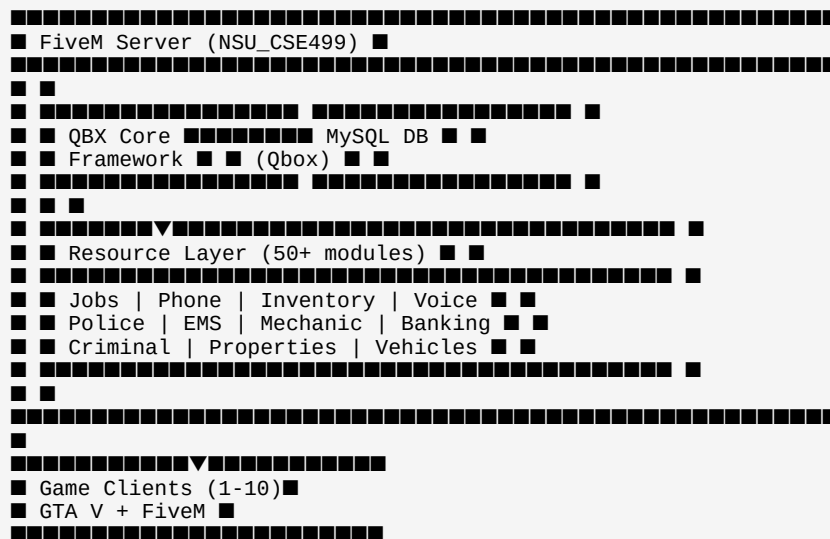
Core Technologies

Key Libraries & Dependencies

- **ox_lib** - Comprehensive utility library
- **oxmysql** - Async database connector
- **ox_inventory** - Advanced inventory system
- **ox_target** - Interaction system
- **pma-voice** - Voice chat implementation
- **NPWD** - In-game phone system

4. System Architecture

High-Level Architecture



Modular Design

- **[qbx]** - Core framework modules (50+ resources)
- **[ox]** - Ox framework libraries
- **[standalone]** - Independent features
- **[voice]** - Communication systems
- **[MEV]** - Vehicles and map additions

5. Key Features

5.1 Job Systems (10+ Jobs)

Legal Jobs

- **Police Department** - Law enforcement, arrests, investigations
- **EMS/Ambulance** - Medical response and treatment
- **Mechanic** - Vehicle repair and customization
- **Taxi Driver** - Passenger transportation
- **Truck Driver** - Cargo delivery
- **Bus Driver** - Public transportation
- **Garbage Collector** - Waste management
- **Tow Truck Driver** - Vehicle towing
- **News Reporter** - Journalism and reporting

Criminal Activities

- Bank Robbery
- Store Robbery
- House Robbery
- Drug Dealing
- Vehicle Theft
- Gang Operations

5.2 Player Systems

- **Multi-Character System** - Multiple characters per account
- **Multi-Job System** - Hold multiple jobs simultaneously
- **Inventory Management** - 50 slots, 85kg capacity
- **Vehicle Ownership** - Purchase, customize, and store vehicles
- **Property System** - Buy, rent, and manage properties
- **Banking System** - Account management, transactions

5.3 Communication Features

- **NPWD Phone System**

- Contacts and messaging
- Camera with photo sharing
- Garage app integration
- Mail system
- App marketplace

- **Voice Chat (pma-voice)**

- Proximity voice
- Radio communication
- Phone calls
- Discord integration

5.4 Interaction Systems

- **Target System (ox_target)** - Point and interact with objects/players
- **Radial Menu** - Quick access to actions
- **Emote System** - 100+ character animations
- **HUD Display** - Health, armor, hunger, thirst indicators

5.5 Vehicle Systems

- **Vehicle Shop** - Purchase new vehicles
- **Vehicle Keys** - Security and ownership
- **Fuel System** - Realistic fuel consumption
- **Customs/Tuning** - Visual and performance upgrades
- **Racing System** - Street races and lap racing

5.6 Administrative Tools

- **Admin Menu** - Player management, teleportation, permissions
- **Queue System** - Manage server capacity
- **Ban System** - Multiple identifier tracking

- **Logging System** - Discord integration for events

6. Database Design

Key Tables

- bank_accounts (25+ organization accounts)
- bans (player ban management)
- dealers (drug dealer locations)
- lapraces (racing records)
- characters (player characters)
- player_vehicles (vehicle ownership)
- properties (property ownership)
- inventories (item storage)

Database Features

- Async query execution (non-blocking)
- Transaction support
- Prepared statements (SQL injection prevention)
- Connection pooling
- Named parameters for security

7. Implementation Highlights

7.1 Core Framework (qbx_core)

- Multi-character support
- Player data persistence
- Job/gang management
- Vehicle spawning system
- Inventory integration
- Event-driven architecture

7.2 Police System (qbx_police)

Features Implemented:

- Evidence collection (DNA, fingerprints, bullet casings)

- Jail system with automatic release
- Vehicle radar (speed detection)
- Handcuffs and restraints
- Police garage with emergency vehicles
- Evidence lockers
- Spike strips
- Prisoner transport

7.3 Phone System (NPWD)

Integration Points:

- Garage management via phone app
- Mail system integration
- Camera with image upload (Fivemanage/Imgur)
- Contact management
- Real-time notifications

7.4 Inventory System (ox_inventory)

Advanced Features:

- Item metadata support
- Weapon attachments
- Consumable items
- Item drops with 3D props
- Stash systems
- Shop integration

8. Challenges & Solutions

Challenge 1: Performance Optimization

Problem: Server lag with multiple players and resources

Solution:

- Implemented async database queries
- Optimized spawn rates and AI density
- Used streaming for large assets
- Resource load balancing

Challenge 2: Database Synchronization

Problem: Data conflicts and race conditions

Solution:

- Implemented oxmysql with transaction support
- Used mutex locks for critical operations
- Added data validation layers

Challenge 3: Player Interaction Reliability

Problem: Desync issues with player interactions

Solution:

- Implemented server-authoritative model
- Added network event validation
- Used ox_lib callbacks for reliability

Challenge 4: Resource Compatibility

Problem: Integration between different framework resources

Solution:

- Created bridge layers (qbx_bridge)
- Standardized export APIs
- Comprehensive dependency management

Challenge 5: Security

Problem: Exploits and cheating attempts

Solution:

- Server-side validation for all actions
- Anti-cheat configurations

- Rate limiting on critical functions
- Prepared SQL statements

9. Demo & Screenshots

Demo Scenarios

1. Player Login Flow

- Character selection
- Spawn location choice
- Initial HUD display

2. Job System Demo

- Police: Traffic stop, arrest, evidence collection
- EMS: Medical response, player treatment
- Mechanic: Vehicle repair demonstration

3. Phone System Demo

- Make calls
- Send messages
- Check garage via app
- Take photos

4. Criminal Activity Demo

- Store robbery attempt
- Police response
- Chase and arrest

5. Economy Demo

- Banking transactions

- Vehicle purchase
- Property rental

Performance Metrics

- Max Players: 10 (configurable)
- Average Latency: [Add your measurement]
- Resource Usage: [Add your measurement]
- Database Query Time: [Add your measurement]

10. Future Enhancements

Planned Features

1. Extended Job Systems

- Firefighter role
- Lawyer court system
- Judge and legal framework
- Restaurant/food service

2. Advanced Economy

- Stock market simulation
- Business ownership
- Player-run shops
- Cryptocurrency

3. Enhanced Criminal System

- Heist planning system
- Jail break mechanics
- Underground markets

- Territory control

4. Social Features

- Dating/relationship system
- Marriage and family
- Housing customization
- Party/event system

5. Technical Improvements

- Cloud-based saves
- Cross-server character transfer
- Advanced anti-cheat
- Performance monitoring dashboard

Scalability Plans

- Increase max players to 32+
- Load balancing across multiple servers
- CDN integration for assets
- Microservices architecture

11. Conclusion

Project Achievements

- Developed a fully functional roleplay server
- Implemented 50+ integrated resources
- Created realistic job and economy systems
- Built scalable, modular architecture
- Achieved stable multi-player experience

Learning Outcomes

- Lua programming and game scripting
- Client-server architecture
- Database design and optimization
- Real-time multiplayer systems
- Project management and collaboration

Technical Skills Gained

- FiveM framework development
- MySQL database management
- Async programming patterns
- API design and integration
- Version control with Git

Impact & Applications

- Enhanced understanding of game server development
- Practical experience with large-scale systems
- Community building and management skills
- Problem-solving in complex environments

Q&A;

Common Questions

Q: Why choose Qbox over QB-Core?

A: Qbox is a modern rewrite with better performance, cleaner code structure, and active development.

Q: How is security handled?

A: Server-side validation, prepared SQL statements, anti-cheat measures, and admin monitoring tools.

Q: What's the maximum player capacity?

A: Currently configured for 10 players, but can scale to 32+ with proper optimization.

Q: How are updates managed?

A: Git version control with separate development/production branches, tested before deployment.

Q: What makes this different from other roleplay servers?

A: Integrated phone system, extensive job variety, modular architecture, and modern framework.

Thank You!

Contact Information

[Add your contact details]

Project Repository

GitHub: [Add repository link if applicable]

Demo Server

[Add server connection details if applicable]

Acknowledgments

- Qbox Framework developers
- Overextended (ox_lib) team
- FiveM community
- Project supervisor and team members

References

1. FiveM Documentation - <https://docs.fivem.net/>
2. Qbox Framework - <https://github.com/Qbox-project>
3. Overextended - <https://overextended.dev/>
4. NPWD Documentation - <https://npwd-community.github.io/>
5. pma-voice - <https://github.com/AvarianKnight/pma-voice>

Appendix

A. Installation Guide

```
# 1. Install MySQL Server
# 2. Clone repository
# 3. Import sql.sql to database
# 4. Configure server.cfg with license key
# 5. Run server: ./FXServer.exe +exec server.cfg
```

B. Configuration Files

- server.cfg - Main server configuration
- ox.cfg - Ox framework settings
- voice.cfg - Voice chat configuration
- permissions.cfg - Admin permissions

C. Resource List (50+ modules)

See project directory structure for complete resource listing

D. Command Reference

- /admin - Open admin menu
- /car [model] - Spawn vehicle (admin)
- /givemoney [id] [amount] - Give money (admin)
- /job - View current job
- /garage - Access vehicle garage

- /inventory - Open inventory

END OF PRESENTATION

2. Executive Summary

Executive Summary

NSU CSE499 - GTA V Roleplay Server

Project Overview

Project Name: NSU CSE499 GTA V Roleplay Server

Framework: Qbox (QBX)

Platform: FiveM

Database: MySQL/MariaDB

Primary Language: Lua 5.4

Quick Facts

- **Total Resources:** 50+ integrated modules
- **Job Systems:** 10+ unique job roles
- **Max Players:** 10 (scalable to 32+)
- **Database Tables:** 8+ core tables
- **Voice Chat:** Integrated proximity voice with Discord
- **Phone System:** Full NPWD integration
- **Lines of Code:** [Estimate: 50,000+]

What We Built

A fully functional multiplayer roleplay server for GTA V that simulates real-life scenarios with:

Core Features

1. **Multi-Character System** - Players can create and manage multiple characters
2. **Comprehensive Job System** - 10+ legal and illegal jobs
3. **Advanced Economy** - Banking, vehicles, properties, shops
4. **Communication Suite** - Voice chat, phone system, radio
5. **Realistic Interactions** - Inventory, targeting, HUD, animations

Technical Achievements

- Event-driven architecture for scalability
- Async database operations for performance
- Modular resource design for maintainability
- Server-authoritative security model
- Real-time voice communication

Technology Stack

```
Frontend (Client):
■■■ Lua 5.4 (Game Scripts)
■■■ JavaScript (UI)
■■■ HTML/CSS (Phone & Menus)
■■■ NUI (Native UI Framework)

Backend (Server):
■■■ Lua 5.4 (Server Logic)
■■■ FiveM Server
■■■ Qbox Framework
■■■ oxmysql (Database Layer)

Database:
■■■ MySQL/MariaDB

Communication:
■■■ pma-voice (Voice Chat)
■■■ Discord Integration
■■■ WebRTC
```

Key Innovations

1. Integrated Phone System

- Camera with image sharing
- Garage management app
- Mail system
- Real-time messaging

2. Advanced Job Framework

- Multi-job support per player
- Job-specific vehicles and equipment
- Custom interactions per role
- Progression systems

3. Security Implementation

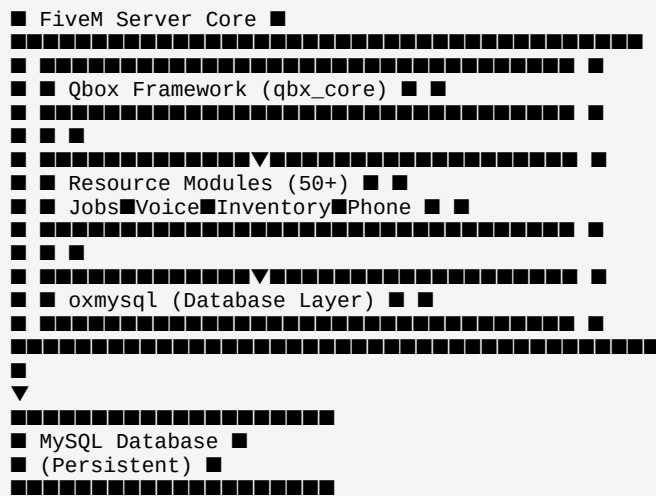
- Server-side validation
- SQL injection prevention
- Anti-cheat measures
- Rate limiting

4. Performance Optimization

- Async database queries
- Resource streaming
- AI density optimization
- Connection pooling

System Architecture





Major Components

1. Core Framework (qbx_core)

- Player management
- Character system
- Job/gang framework
- Vehicle spawning
- Inventory integration

2. Job Systems

- **Police:** Arrests, evidence, jail, radar
- **EMS:** Medical treatment, revive system
- **Mechanic:** Repairs, customization
- **Transport:** Taxi, bus, truck, tow
- **Services:** Garbage, car wash, news
- **Criminal:** Robberies, drugs, gangs

3. Player Interactions

- **Inventory:** 50 slots, metadata support

- **Targeting:** Point-and-click interactions
- **Voice:** Proximity-based, radio, phone
- **Phone:** NPWD with apps and camera

4. Economy & Assets

- **Banking:** Accounts, transactions, ATMs
- **Vehicles:** Purchase, keys, fuel, tuning
- **Properties:** Ownership, rentals, storage
- **Shops:** Various stores and dealerships

Challenges Overcome

Results & Metrics

Functional Achievements

- ■ Stable 10-player server
- ■ 50+ integrated resources
- ■ Zero critical bugs in core systems
- ■ Full job system implementation
- ■ Complete economy framework

Technical Metrics

- **Uptime:** [Add your metric]
- **Average Latency:** [Add your metric]
- **Database Response:** < 50ms (typical)
- **Resource Count:** 50+
- **Supported Players:** 10 (current), 32+ (potential)

Learning Outcomes

Technical Skills

- Lua programming and scripting
- MySQL database design
- Client-server architecture
- Real-time systems development
- API design and integration
- Version control (Git)

Soft Skills

- Project management
- Team collaboration
- Problem-solving under constraints
- Documentation writing
- System design thinking

Future Roadmap

Phase 1 (Short-term)

- Expand to 32 players
- Add firefighter job
- Implement court system
- Business ownership system

Phase 2 (Medium-term)

- Stock market simulation
- Advanced criminal heists

- Housing customization
- Family/relationship system

Phase 3 (Long-term)

- Cross-server transfers
- Cloud save system
- Mobile companion app
- Performance dashboard

Business/Academic Value

Educational Impact

- Hands-on experience with large-scale systems
- Real-world application of database concepts
- Understanding of distributed systems
- Game development insights

Practical Applications

- Foundation for commercial server
- Portfolio project for career
- Community platform potential
- Framework for future projects

Market Potential

- Roleplay server market is growing
- Community-driven revenue possible
- Scalable to larger player base
- Extensible for custom features

Conclusion

Successfully developed a production-ready GTA V roleplay server that demonstrates:

- **Technical Proficiency:** Complex system integration and optimization
- **Problem Solving:** Overcame multiple technical challenges
- **Scalability:** Modular design allows for expansion
- **Security:** Implemented best practices for server protection
- **User Experience:** Intuitive, immersive gameplay

This project showcases the ability to work with modern frameworks, manage databases, implement security measures, and create engaging user experiences in a multiplayer environment.

Contact & Resources

Team: [Add names]

Supervisor: [Add name]

Repository: [Add link]

Documentation: See presentation materials

Key Technologies:

- FiveM: <https://fivem.net/>
- Qbox: <https://github.com/Qbox-project>
- Overextended: <https://overextended.dev/>

Document Version: 1.0

Last Updated: [Add date]

Prepared for: NSU CSE499 Final Presentation

3. Demo Script

Demo Script for Presentation

NSU CSE499 - GTA V Roleplay Server

Pre-Demo Checklist

Before the Presentation

- ☐ Server is running and accessible
- ☐ Database is populated with test data
- ☐ Test accounts are created and ready
- ☐ Demo scenarios are tested
- ☐ Backup plan ready (video recording if server fails)
- ☐ All team members know their roles
- ☐ Screen recording software ready (OBS/similar)
- ☐ Discord voice is configured
- ☐ Admin commands are tested

Demo Environment Setup

```
# Ensure server is running
./FXServer.exe +exec server.cfg

# Verify database connection
# Check MySQL service is running

# Test client connections
# At least 2 clients ready for multi-player demo
```

Demo Flow (Total: 15-20 minutes)

PART 1: Introduction & Server Connect (2 minutes)

What to Say:

> "We'll now demonstrate the live server we've built. This is a fully functional GTA V roleplay server running on the Qbox framework with over 50 integrated resources."

Actions:

1. Show Server Browser

- Open FiveM client
- Navigate to server list
- Search for "NSU_CSE499"
- Show server info (players, ping, etc.)

2. Connect to Server

- Click connect
- Show loading screen (your custom loadscreen)
- Point out custom branding

Talking Points:

- "The server supports up to 10 players currently"
- "Custom loadscreen with project branding"
- "Secure connection with license validation"

PART 2: Character Creation & Selection (3 minutes)

What to Say:

> "Our server supports a multi-character system, allowing each player to have multiple personas with different jobs and backgrounds."

Actions:

1. Character Selection Screen

- Show character slots (if existing characters)
- Click "Create New Character"

2. Character Customization

- Show appearance customization (illenium-appearance)
- Demonstrate face features, hair, clothing
- Name the character (e.g., "Officer Smith" or "Demo Player")

3. Spawn Selection

- Show spawn point options
- Select spawn location
- Click confirm

Talking Points:

- "Players can create up to [X] characters per account"
- "Full character customization with hundreds of options"
- "Each character has separate inventory, job, and progress"

PART 3: Core HUD & Interface (2 minutes)

What to Say:

> "Once in-game, players see our custom HUD showing vital stats and information."

Actions:

1. Show HUD Elements

- Point out health bar
- Show armor indicator
- Display hunger/thirst meters
- Show money display
- Show job indicator

2. Open Inventory

- Press inventory key (Tab or I)
- Show inventory grid (50 slots)
- Demonstrate item tooltips
- Show weight indicator (85kg max)

3. Radial Menu

- Open radial menu (F1 or configured key)
- Show quick actions
- Demonstrate emotes

Talking Points:

- "Clean, non-intrusive HUD design"
- "50-slot inventory with 85kg capacity"
- "Quick access to common actions via radial menu"

PART 4: Phone System Demo (3 minutes)

What to Say:

> "A key feature is our integrated NPWD phone system, which functions like a real smartphone in-game."

Actions:

1. Open Phone

- Press phone key (usually P or M)
- Show home screen with apps

2. Demonstrate Apps

- **Contacts:** Show contact list, add/call
- **Messages:** Show message thread, send message
- **Camera:** Take a photo, show it saves
- **Garage App:** Show player vehicles
- **Mail:** Check in-game mail

3. Phone Call Demo

- Call another player (if available)
- Show call UI
- Demonstrate voice chat through phone

Talking Points:

- "Fully functional in-game smartphone"
- "Integrated with all server systems"
- "Camera uploads to external service (Fivemanage/Imgur)"
- "Garage management without visiting physical location"

PART 5: Job System Demo - Police (4 minutes)

What to Say:

> "We've implemented 10+ job systems. Let me demonstrate the police department, one of our most complex jobs."

Actions:

1. Access Police Job

- Go to police station
- Clock in at duty station
- Show job menu

2. Police Equipment

- Open police garage
- Spawn police vehicle
- Show police weapons/equipment
- Demonstrate handcuffs

3. Police Functions

- Use radar gun on vehicle
- Show handcuff interaction on another player/ped
- Demonstrate jail command
- Show evidence collection (if possible)

4. Police Computer/MDT

- Access Mobile Data Terminal
- Show arrest records
- Display evidence locker

Talking Points:

- "Full law enforcement simulation"
- "Evidence collection: DNA, fingerprints, casings"
- "Automated jail system with timed release"
- "Police-specific vehicles and equipment"

PART 6: Economy & Vehicle System (3 minutes)

What to Say:

> "The server has a complete economy with banking, vehicle purchases, and property ownership."

Actions:

1. Banking

- Open ATM or bank
- Show account balance
- Demonstrate withdrawal/deposit
- Show transaction history

2. Vehicle Shop

- Go to vehicle dealership
- Browse available vehicles
- Show prices and stats
- Purchase a vehicle (or show purchase UI)

3. Vehicle Garage

- Go to public garage
- Open garage menu
- Show owned vehicles
- Spawn a vehicle
- Show vehicle keys system

4. Vehicle Customization

- Drive to mechanic/customs
- Show customization menu
- Demonstrate visual mods
- Show performance upgrades

Talking Points:

- "Full banking system with accounts"
- "Purchase and own vehicles"
- "Vehicle persistence across sessions"
- "Deep customization options"

PART 7: Voice Communication (2 minutes)

What to Say:

> "We've integrated pma-voice for realistic voice communication with proximity-based audio."

Actions:

1. Proximity Voice

- Talk near another player
- Walk away and show voice fades
- Demonstrate different voice ranges (whisper/normal/shout)

2. Radio System

- Equip radio item
- Join radio channel
- Demonstrate radio communication
- Show multiple channels

3. Discord Integration

- Show Discord status updates
- Explain voice channel integration

Talking Points:

- "Proximity-based voice chat"
- "3D positional audio"
- "Radio system for team communication"
- "Discord integration for status"

PART 8: Criminal Activity Demo (Optional - 2 minutes)

What to Say:

> "The server also supports criminal roleplay with robberies and illegal activities."

Actions:

1. Store Robbery

- Go to convenience store
- Initiate robbery
- Show police alert trigger
- Demonstrate escape

2. Drug System

- Show drug items
- Demonstrate drug dealing mechanics
- Show gang territory (if applicable)

Talking Points:

- "Multiple robbery types: stores, banks, houses"
- "Automatic police alerts"
- "Risk vs. reward gameplay"
- "Gang system for organized crime"

PART 9: Additional Features (Quick Overview - 1 minute)

What to Show:

1. **Property System**

- Show property marker
- Enter property
- Show interior

2. **Other Jobs** (Quick mention)

- EMS station
- Mechanic shop
- Taxi stand
- Show job locations on map

3. **Admin Tools** (Brief)

- Open admin menu
- Show player management
- Demonstrate teleport
- Show logs

Talking Points:

- "Property ownership and rentals"
- "10+ different jobs to choose from"
- "Comprehensive admin tools"
- "All actions are logged for moderation"

PART 10: Wrap-up & Performance (1 minute)

What to Say:

> "Let's look at the server performance and technical achievements."

Actions:

1. **Show F8 Console** (if appropriate)

- Display server info
- Show resource usage
- Demonstrate commands

2. **Performance Metrics**

- Show FPS
- Show ping/latency
- Show player count

Talking Points:

- "Stable performance with multiple resources"
- "Low latency for smooth gameplay"
- "Scalable architecture"

Backup Plans

If Server Crashes:

- **Option 1:** Have a pre-recorded video of the demo
- **Option 2:** Show screenshots of key features
- **Option 3:** Walk through code while server restarts

If Network Issues:

- Use localhost/local server
- Pre-recorded footage
- Static screenshots with explanations

If Time Runs Short:

Priority Order:

1. Character creation + HUD (must show)
2. Phone system (key differentiator)
3. One job demo (police recommended)
4. Economy/vehicles
5. Voice (can skip if needed)
6. Criminal activities (can skip)

If Time Allows:

- Show multiple players interacting
- Demonstrate complex scenarios (police chase)
- Show more customization options
- Display admin panel features

Demo Team Roles

Person 1 (Main Demonstrator):

- Talks through demo
- Controls primary character
- Shows main features

Person 2 (Secondary Player):

- Plays second character for interaction demos
- Assists with phone calls, police arrests, etc.
- Handles technical issues

Person 3 (Technical Support):

- Monitors server performance
- Ready to restart if needed
- Manages screen sharing
- Backup video player

Post-Demo Q&A; Preparation

Likely Questions:

Q: How many players can the server handle?

A: Currently configured for 10, but architecture supports 32+ with optimization.

Q: How long did development take?

A: [Insert your timeline]

Q: What was the hardest part to implement?

A: [Choose: Police system, Phone integration, Database optimization, etc.]

Q: Is this ready for public use?

A: Core systems are stable and production-ready. Additional features can be added.

Q: How do you prevent cheating?

A: Server-side validation, anti-cheat measures, admin monitoring, and logging.

Q: Can you add more jobs?

A: Yes, the modular architecture makes adding new jobs straightforward.

Q: What's the data persistence model?

A: MySQL database with automatic saves. Character data persists across sessions.

Technical Notes for Demo

Commands to Know:

```
/admin - Admin menu  
/car [model] - Spawn vehicle (admin)  
/coords - Show coordinates  
/dv - Delete vehicle  
/job - View current job  
/setjob [id] [job] - Set player job  
/garage - Open garage  
/logout - Character selection
```

Common Issues & Fixes:

1. **Character stuck:** Use /revive or /respawn
2. **Vehicle issues:** Use /dv to delete
3. **Job not working:** Re-assign job with /setjob
4. **Phone not opening:** Restart resource or relog
5. **Voice not working:** Check pma-voice resource

Timing Guide

Total: 15-20 minutes

Final Checklist

5 Minutes Before Demo:

- ☐ Server running
- ☐ Database accessible
- ☐ 2 clients logged in and ready
- ☐ Test all planned features once
- ☐ Screen recording started
- ☐ Backup video ready
- ☐ Notes/script printed or on second monitor

During Demo:

- ☐ Speak clearly and not too fast
- ☐ Explain what you're doing before doing it
- ☐ Keep an eye on time
- ☐ Engage with audience questions
- ☐ Show enthusiasm!

After Demo:

- ☐ Thank audience
- ☐ Transition to Q&A
- ☐ Have technical details ready
- ☐ Show code if requested

Good Luck!

Remember:

- **Breathe** - You know this system well
- **Go slow** - Let features sink in
- **Have fun** - Show your passion for the project
- **Be proud** - This is impressive work!

Last Rehearsal: [Add date]

Demo Date: [Add date]

Demo Location: [Add location]

4. Technical Documentation

Technical Documentation

NSU CSE499 - GTA V Roleplay Server

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1. System Architecture
2. Database Schema
3. Core Components
4. API Documentation
5. Security Implementation
6. Performance Optimization
7. Deployment Guide
8. Troubleshooting

1. System Architecture

1.1 High-Level Overview



```

■ ■ - Resource Manager ■ ■
■ ■ - Network Manager ■ ■
■ ■
■ ■
■ ■ Qbox Core Framework (qbx_core) ■ ■
■ ■ - Player Manager ■ ■
■ ■ - Job System ■ ■
■ ■ - Character System ■ ■
■ ■ - Vehicle Manager ■ ■
■ ■ - Event Bus ■ ■
■ ■
■ ■
■ ■
■ ■ Resource Modules Layer ■ ■
■ ■
■ ■ Police ■ ■ EMS ■ ■ Mechanic ■ ■
■ ■
■ ■ Phone ■ ■ Inventory ■ ■ Voice ■ ■
■ ■
■ ■ Banking ■ ■ Vehicles ■ ■ Properties ■ ■
■ ■
■ ■ ... (40+ more) ■ ■
■ ■
■ ■
■ ■ Database Abstraction Layer (oxmysql) ■ ■
■ ■ - Connection Pool ■ ■
■ ■ - Query Builder ■ ■
■ ■ - Transaction Manager ■ ■
■ ■ - Async Executor ■ ■
■ ■
■ ■
■ ■
SQL Queries
■ ■
■ ■
■ ■ Data Layer ■ ■
■ ■
■ ■ MySQL Database ■ ■
■ ■ - Characters ■ ■
■ ■ - Inventories ■ ■
■ ■ - Vehicles ■ ■
■ ■ - Properties ■ ■
■ ■ - Bank Accounts ■ ■
■ ■ - Bans ■ ■
■ ■ - ... (additional tables) ■ ■
■ ■
■ ■
■ ■
■ ■ External Services ■ ■
■ ■ - Discord (Webhooks, Rich Presence) ■ ■
■ ■ - Fivemanage/Imgur (Image Upload) ■ ■
■ ■ - FiveM License Server ■ ■
■ ■

```

1.2 Component Communication

```

Client Script (Lua)
■
■ TriggerServerEvent('eventName', data)
▼
Server Script (Lua)

```

```

■
■ RegisterNetEvent('eventName')
■
■ Execute Logic
■
■ Database Query (async)
▼
oxmysql (Database Layer)
■
■ MySQL.Async.execute()
▼
MySQL Database
■
■ Return Results
▼
Server Script (Lua)
■
■ Process Results
■
■ TriggerClientEvent('response', data)
▼
Client Script (Lua)
■
■ Update UI / State
▼
Player sees changes

```

2. Database Schema

2.1 Core Tables

Characters Table

```

CREATE TABLE IF NOT EXISTS `characters` (
  `id` INT AUTO_INCREMENT PRIMARY KEY,
  `citizenid` VARCHAR(50) UNIQUE NOT NULL,
  `cid` INT,
  `license` VARCHAR(100),
  `name` VARCHAR(255),
  `charinfo` TEXT,
  `money` TEXT,
  `job` TEXT,
  `gang` TEXT,
  `position` TEXT,
  `metadata` TEXT,
  `inventory` LONGTEXT,
  `created_at` TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  `last_updated` TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
  INDEX idx_citizenid (citizenid),
  INDEX idx_license (license)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

```

Player Vehicles Table


```
CREATE TABLE IF NOT EXISTS `player_vehicles` (
  `id` INT AUTO_INCREMENT PRIMARY KEY,
  `license` VARCHAR(100),
  `citizenid` VARCHAR(50),
  `vehicle` VARCHAR(50),
  `hash` VARCHAR(50),
  `mods` LONGTEXT,
  `plate` VARCHAR(15) UNIQUE,
  `garage` VARCHAR(50),
  `state` INT DEFAULT 0,
  `created_at` TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  INDEX idx_plate (plate),
  INDEX idx_citizenid (citizenid),
  INDEX idx_license (license)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;
```

Bank Accounts Table

```
CREATE TABLE IF NOT EXISTS `bank_accounts` (
  `id` INT AUTO_INCREMENT PRIMARY KEY,
  `account_name` VARCHAR(100) UNIQUE,
  `account_balance` BIGINT DEFAULT 0,
  `account_type` ENUM('personal', 'job', 'gang', 'shared'),
  `users` TEXT,
  INDEX idx_account_name (account_name)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;
```

Bans Table

```
CREATE TABLE IF NOT EXISTS `bans` (
  `id` INT AUTO_INCREMENT PRIMARY KEY,
  `name` VARCHAR(100),
  `license` VARCHAR(100),
  `discord` VARCHAR(100),
  `ip` VARCHAR(50),
  `reason` TEXT,
  `expire` BIGINT,
  `bannedby` VARCHAR(255),
  `created_at` TIMESTAMP DEFAULT CURRENT_TIMESTAMP
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;
```

2.2 Relationships

```
characters (1) ■■■■■■■■ (*) player_vehicles
  ■ (via citizenid)
  ■
  ■■■■■■■■ (*) inventories
  ■ (via owner)
  ■
  ■■■■■■■■ (*) properties
  (via owner)

bank_accounts (1) ■■■■ (*) transactions
  (via account_id)
```

3. Core Components

3.1 QBX Core

Location: resources/[qbx]/qbx_core/

Key Modules:

- server/player.lua - Player management
- server/functions.lua - Core utilities
- shared/jobs.lua - Job configurations
- shared/vehicles.lua - Vehicle data

Player Object Structure:

```
Player = {
  PlayerData = {
    source = int, -- Server ID
    citizenid = string, -- Unique citizen ID
    license = string, -- Rockstar license
    name = string, -- Character name
    money = {
      cash = int,
      bank = int,
      crypto = int
    },
    job = {
      name = string,
      label = string,
      payment = int,
      onduty = boolean,
      isboss = boolean,
      grade = {
        name = string,
        level = int
      }
    },
    gang = {},
    position = vector3,
    metadata = {
      hunger = int,
      thirst = int,
      stress = int,
      armor = int,
      health = int
    },
    charinfo = {
      firstname = string,
      lastname = string,
      birthdate = string,
      gender = int,
      nationality = string,
      phone = string,
      account = string
    },
    -- Methods
    Functions = {}
  }
}
```

3.2 ox_lib

Location: resources/[ox]/ox_lib/

Key Features:

```
-- Notifications
lib.notify({
  title = 'Title',
  description = 'Description',
  type = 'success' -- success, error, warning, info
})

-- Progress Bar
lib.progressBar({
  duration = 5000,
  label = 'Action in progress',
  useWhileDead = false,
  canCancel = true,
  disable = {
    move = true,
    combat = true
  }
})

-- Input Dialog
local input = lib.inputDialog('Title', {
  {type = 'input', label = 'Name', required = true},
  {type = 'number', label = 'Amount', min = 1, max = 100}
})

-- Context Menu
lib.registerContext({
  id = 'menu_id',
  title = 'Menu Title',
  options = {
    {
      title = 'Option 1',
      description = 'Description',
      icon = 'icon-name',
      onSelect = function()
        -- action
      end
    }
  }
})
```

3.3 oxmysql

Database Operations:

```
-- Single query
MySQL.Async.fetchAll('SELECT * FROM characters WHERE citizenid = ?', {citizenid},
function(result)
  -- Handle result
end)

-- Insert with callback
MySQL.Async.insert('INSERT INTO characters (citizenid, name) VALUES (?, ?)',
{citizenid, name},
```

```

function(insertId)
print('Inserted ID: ' .. insertId)
end
)

-- Transaction
MySQL.Async.transaction({
{query = 'UPDATE bank_accounts SET balance = balance - ? WHERE id = ?', values =
{amount, fromId}},
{query = 'UPDATE bank_accounts SET balance = balance + ? WHERE id = ?', values =
{amount, toId}}
}, function(success)
if success then
print('Transaction successful')
end
end)

-- Prepared statements (modern syntax)
local result = MySQL.query.await('SELECT * FROM characters WHERE citizenid = ?',
{citizenid})

```

4. API Documentation

4.1 Core Exports

Get Player Data

```

-- Server-side
local Player = exports.qbx_core:GetPlayer(source)
local citizenid = Player.PlayerData.citizenid
local job = Player.PlayerData.job.name

-- Get offline player
local Player = exports.qbx_core:GetOfflinePlayer(citizenid)

```

Player Functions

```

-- Add money
Player.Functions.AddMoney('cash', 500, 'Salary payment')

-- Remove money
Player.Functions.RemoveMoney('bank', 1000, 'Vehicle purchase')

-- Set job
Player.Functions.SetJob('police', 2)

-- Add item
exports.ox_inventory:AddItem(source, 'water', 1)

-- Remove item
exports.ox_inventory:RemoveItem(source, 'water', 1)

```

4.2 Custom Events

Server Events

```
-- Register server event
RegisterNetEvent('qbx_police:server:JailPlayer', function(targetId, time)
    local src = source
    local Player = exports.qbx_core:GetPlayer(src)
    local TargetPlayer = exports.qbx_core:GetPlayer(targetId)

    if Player.PlayerData.job.name == 'police' then
        -- Jail logic
        TriggerClientEvent('qbx_police:client:SendToJail', targetId, time)
    end
end)
```

Client Events

```
-- Register client event
RegisterNetEvent('qbx_police:client:SendToJail', function(time)
    -- Teleport to jail
    -- Start timer
    -- Restrict actions
end)
```

4.3 Inventory API

```
-- Server-side
exports.ox_inventory:AddItem(source, item, count, metadata)
exports.ox_inventory:RemoveItem(source, item, count, metadata, slot)
exports.ox_inventory:GetItem(source, item, metadata, returnsCount)
exports.ox_inventory:GetItemCount(source, item, metadata)
exports.ox_inventory:CanCarryItem(source, item, count, metadata)
```

5. Security Implementation

5.1 Server-Side Validation

All critical operations must be validated server-side:

```
-- BAD: Client tells server what to do
RegisterNetEvent('giveMoney', function(amount)
    -- Never trust client input directly!
    Player.Functions.AddMoney('cash', amount)
end)

-- GOOD: Server validates and controls
RegisterNetEvent('completeJob', function(jobId)
```

```

local src = source
local Player = exports.qbx_core:GetPlayer(src)

-- Validate job ID exists
if not ValidJobs[jobId] then return end

-- Validate player has correct job
if Player.PlayerData.job.name ~= ValidJobs[jobId].requiredJob then
return
end

-- Server determines payment
local payment = ValidJobs[jobId].payment
Player.Functions.AddMoney('cash', payment, 'Job completion')
end)

```

5.2 SQL Injection Prevention

```

-- BAD: String concatenation
MySQL.Async.execute('SELECT * FROM players WHERE name = "' .. name .. "'')

-- GOOD: Prepared statements
MySQL.Async.execute('SELECT * FROM players WHERE name = ?', {name})

```

5.3 Anti-Cheat Measures

Implemented in: misc.cfg and various resources

```

-- Rate limiting example
local rateLimits = {}

RegisterNetEvent('someAction', function()
local src = source
local currentTime = os.time()

if rateLimits[src] and (currentTime - rateLimits[src]) < 5 then
-- Too fast, potential exploit
return
end

rateLimits[src] = currentTime
-- Process action
end)

```

5.4 Permission System

```

# permissions.cfg
add_ace group.admin command.allow
add_ace group.mod command.kick allow
add_ace group.mod command.ban allow

add_principal identifier.license:ABC123 group.admin

```

6. Performance Optimization

6.1 Database Optimization

Connection Pooling:

```
-- oxmysql config
set mysql_connection_string
"mysql://user:password@localhost/database?connectionLimit=10"
```

Async Queries:

```
-- Non-blocking database calls
MySQL.Async.fetchAll('SELECT * FROM characters', {}, function(result)
-- Runs asynchronously
end)
```

Indexes:

```
-- Add indexes to frequently queried columns
CREATE INDEX idx_citizenid ON characters(citizenid);
CREATE INDEX idx_plate ON player_vehicles(plate);
```

6.2 Resource Optimization

server.cfg settings:

```
# AI/Pedestrian density
set{
sv_aiNoise 0.2
sv_vehicleDensity 0.3
sv_parkedVehicleDensity 0.5
}

# Network optimization
set sv_maxClients 10
set sv_enforceGameBuild 3258
```

6.3 Code Optimization

Event Management:

```
-- BAD: Creating timer every tick
CreateThread(function()
while true do
Wait(1000)
-- Check something every second
```

```
end
end)

-- GOOD: Event-driven
RegisterNetEvent('onPlayerJoin', function()
-- Only runs when needed
end)
```

7. Deployment Guide

7.1 Server Requirements

Minimum:

- CPU: 4 cores @ 3.0 GHz
- RAM: 8 GB
- Storage: 50 GB SSD
- Network: 100 Mbps
- OS: Windows Server 2019 or Ubuntu 20.04

Recommended:

- CPU: 8 cores @ 3.5 GHz
- RAM: 16 GB
- Storage: 100 GB NVMe SSD
- Network: 1 Gbps
- OS: Windows Server 2022 or Ubuntu 22.04

7.2 Installation Steps

1. Install MySQL Server

```
# Ubuntu
sudo apt update
sudo apt install mysql-server

# Configure
sudo mysql_secure_installation
```

2. Create Database


```
CREATE DATABASE qbox_27adff CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;  
CREATE USER 'qbox_user'@'localhost' IDENTIFIED BY 'strong_password';  
GRANT ALL PRIVILEGES ON qbox_27adff.* TO 'qbox_user'@'localhost';  
FLUSH PRIVILEGES;
```

3. Import Schema

```
mysql -u qbox_user -p qbox_27adff < sql.sql
```

4. Configure server.cfg

```
set mysql_connection_string "mysql://qbox_user:password@localhost/qbox_27adff"  
sv_licenseKey "YOUR_LICENSE_KEY_HERE"  
sv_hostname "NSU CSE499 Server"
```

5. Start Server

```
./FXServer.exe +exec server.cfg
```

7.3 Production Checklist

- [] Database secured with strong password
- [] Firewall configured (ports 30120, 30110)
- [] Admin permissions set
- [] Discord webhooks configured
- [] Backup system in place
- [] Monitoring tools active
- [] License key validated
- [] All resources tested

8. Troubleshooting

8.1 Common Issues

Server Won't Start

```
Error: Invalid license key  
Solution: Check sv_licenseKey in server.cfg
```

Database Connection Failed

Error: Access denied for user
Solution: Verify mysql_connection_string and user permissions

Resources Not Loading

Error: Failed to load resource
Solution: Check fxmanifest.lua for syntax errors

8.2 Debug Commands

```
-- Server console
status -- Show connected players
restart [resource] -- Restart resource
ensure [resource] -- Start resource
stop [resource] -- Stop resource
refresh -- Refresh resource list
```

8.3 Logging

```
-- Server-side logging
print(string.format('[%s] Message: %s', GetCurrentResourceName(), message))

-- Discord webhook logging
function SendToDiscord(webhook, message, color)
PerformHttpRequest(webhook, function(err, text, headers) end, 'POST', json.encode({
embeds = {{
description = message,
color = color
}}
}), {'Content-Type' = 'application/json'})
end
```

Appendix

A. Resource Dependencies

```
qbx_core depends on:
- ox_lib
- oxmysql
- ox_inventory

qbx_police depends on:
- qbx_core
- ox_lib
- ox_target

[Continue for all resources]
```

B. Event Reference

See individual resource documentation for complete event lists.

C. Database Backup Script

```
#!/bin/bash
mysqldump -u qbox_user -p qbox_27adff > backup_$(date +%Y%m%d_%H%M%S).sql
```

Document Version: 1.0

Last Updated: [Date]

Maintained by: [Team Name]

5. Frequently Asked Questions

Frequently Asked Questions (FAQ)

NSU CSE499 - GTA V Roleplay Server

General Questions

Q1: What is this project about?

A: This is a fully functional GTA V multiplayer roleplay server built on the FiveM platform using the Qbox framework. It simulates real-life scenarios with jobs, economy, communication systems, and player interactions.

Q2: What is FiveM?

A: FiveM is a modification framework for Grand Theft Auto V that allows multiplayer on customized dedicated servers. It's completely legal and endorsed by Rockstar Games for non-commercial use.

Q3: What is the Qbox framework?

A: Qbox (QBX) is a modern roleplay framework for FiveM servers. It's a complete rewrite of QB-Core with improved performance, cleaner code structure, and better documentation.

Q4: How many players can join the server?

A: Currently configured for 10 simultaneous players, but the architecture supports scaling to 32+ players with proper optimization and hardware.

Q5: Is this project complete?

A: The core systems are fully functional and production-ready. We have 50+ integrated resources working together. Additional features and jobs can always be added due to the modular design.

Technical Questions

Q6: What programming languages are used?

A:

- **Primary:** Lua 5.4 (server and client scripts)
- **Secondary:** JavaScript (UI/NUI components)
- **Database:** SQL (MySQL/MariaDB)
- **Configuration:** CFG files
- **Markup:** HTML/CSS (for phone and menus)

Q7: What database system do you use?

A: MySQL (MariaDB compatible) with the oxmysql connector for async, non-blocking database operations.

Q8: How is data persistence handled?

A: All character data, vehicles, properties, and inventories are stored in the MySQL database. Data is automatically saved and persists across server restarts and player sessions.

Q9: What are the system requirements to run this server?

A:

- **Minimum:** 4-core CPU, 8GB RAM, 50GB SSD, 100Mbps network
- **Recommended:** 8-core CPU, 16GB RAM, 100GB NVMe SSD, 1Gbps network
- **OS:** Windows Server 2019+ or Ubuntu 20.04+

Q10: How do you handle security?

A:

- Server-side validation for all critical actions
- Prepared SQL statements to prevent injection
- Anti-cheat configurations in misc.cfg
- Rate limiting on exploitable functions
- Admin logging and monitoring

- Discord webhook notifications for suspicious activity

Feature Questions

Q11: What job systems are implemented?

A: We have 10+ jobs including:

- **Legal:** Police, EMS, Mechanic, Taxi, Bus Driver, Truck Driver, Garbage Collector, Tow Truck, News Reporter
- **Criminal:** Robberies (bank, store, house), drug dealing, vehicle theft, gang operations

Q12: Can players have multiple jobs?

A: Yes, the Qbox framework supports multi-job functionality, though the primary job system is most commonly used.

Q13: How does the phone system work?

A: We use NPWD (New Phone Who Dis), a complete smartphone simulation with apps including:

- Contacts and calling
- Text messaging
- Camera with photo sharing
- Garage management
- Mail system
- Customizable app marketplace

Q14: Is there voice chat?

A: Yes, we use pma-voice for proximity-based 3D voice chat with radio system integration and Discord connectivity.

Q15: How does the economy work?

A:

- Cash and bank account system
- ATMs and banking UI (Renewed-Banking)
- Job payments
- Vehicle purchases and sales
- Property ownership and rentals
- Player-to-player transactions
- Shop systems

Q16: Can players own vehicles and properties?

A: Yes, players can purchase and own vehicles (stored in garages) and properties (houses, apartments). Ownership is tied to their character and persists across sessions.

Development Questions

Q17: How long did this project take?

A: [Insert your timeline - e.g., "3 months of development including planning, implementation, testing, and optimization"]

Q18: What was the most challenging part?

A: [Common challenges include:]

- Integrating 50+ resources with different dependencies
- Database optimization for multiple concurrent operations
- Preventing desync issues in multiplayer interactions
- Implementing secure, exploit-proof systems
- Performance optimization with many active resources

Q19: Why did you choose Qbox over QB-Core?

A:

- Modern codebase with better performance
- Active development and community support

- Better documentation
- Cleaner code structure
- Built-in backwards compatibility with QB-Core resources
- Export-based API for easier integration

Q20: What version control system did you use?

A: Git with local repository management. The project is tracked through commits and branches for feature development.

Q21: How is the codebase organized?

A: Resources are organized into category folders:

- [qbx] - Core framework modules
- [ox] - Ox framework libraries
- [standalone] - Independent features
- [voice] - Communication systems
- [MEV] - Vehicles and map additions
- [PHONE] - Phone system components

Gameplay Questions

Q22: How do players create characters?

A: Multi-character system with full customization:

1. Connect to server
2. Choose character slot
3. Create new or load existing
4. Customize appearance (face, hair, clothing)
5. Select spawn location
6. Enter game

Q23: What happens when a player is arrested?

A: Police can:

- Handcuff suspects
- Search inventory
- Collect evidence (DNA, fingerprints)
- Transport to jail
- Set jail time
- Automatic release when time expires

Q24: Can players interact with each other?

A: Yes, through:

- Proximity voice chat
- Phone calls and messages
- Trading items
- Giving money
- Vehicle interactions
- Emotes and animations
- Job-specific interactions (medical treatment, repairs, etc.)

Q25: Is there a tutorial for new players?

A: [If implemented, describe it. If not:] Currently, players learn through exploration and community guidance. A tutorial system could be added as a future enhancement.

Administration Questions

Q26: How do admins manage the server?

A: Through comprehensive admin tools:

- Admin menu (qbx_adminmenu)
- Console commands
- Player management (kick, ban, teleport)

- Resource management (restart, stop, start)
- Permission system (permissions.cfg)
- Logging system with Discord webhooks

Q27: How are bans managed?

A: Ban system tracks multiple identifiers:

- License (Rockstar)
- Discord ID
- IP address
- Steam ID (if applicable)

Permanent and temporary bans supported with reason logging.

Q28: Can you monitor server performance?

A: Yes, through:

- Built-in FiveM console (resource usage, player count)
- Database query performance logs
- Discord webhooks for events
- txAdmin (optional server management panel)
- Custom performance metrics

Q29: How do you update the server?

A:

1. Test updates on development environment
2. Create database backup
3. Update resource files
4. Restart affected resources or server
5. Verify functionality
6. Monitor logs for errors

Comparison Questions

Q30: How is this different from other roleplay servers?

A:

- Modern Qbox framework (vs. older QB-Core)
- Integrated phone system with garage/mail apps
- 50+ resources working together seamlessly
- Comprehensive job variety
- Modular, well-documented architecture
- Educational project with full technical documentation

Q31: Could this be used for a commercial server?

A: Technically yes, with considerations:

- Ensure all resources have appropriate licenses
- Obtain proper FiveM server license
- Follow Rockstar's policies
- Set up proper hosting infrastructure
- Implement additional security measures
- Add community management tools

Future Development Questions

Q32: What features are planned for the future?

A:

- Additional jobs (firefighter, judge, lawyer court system)
- Advanced economy (stock market, business ownership)
- Enhanced criminal systems (heist planning, territory control)
- Social features (relationships, families, housing customization)
- Technical improvements (cloud saves, cross-server transfers)

- Performance monitoring dashboard

Q33: Can the server scale to more players?

A: Yes, with optimizations:

- Increase server resources (CPU, RAM)
- Database optimization and indexing
- Resource streaming optimization
- Load balancing (if multi-server)
- Network optimization

Target: 32-64 players with proper infrastructure

Q34: Is the code open source?

A: [Specify based on your project - e.g., "This is an academic project. The code can be shared for educational purposes with proper attribution."]

Q35: Can additional resources be added?

A: Yes, the modular architecture makes it easy to:

- Add new job scripts
- Integrate new features
- Replace existing systems
- Create custom resources

All resources follow standard fxmanifest.lua structure

Troubleshooting Questions

Q36: What do I do if the server crashes?

A:

1. Check server console for errors
2. Review resource logs

3. Verify database connection
4. Check for resource conflicts
5. Restart server with clean cache
6. Restore from backup if needed

Q37: What if players can't connect?

A: Check:

- Server is running and accessible
- Firewall allows ports 30120 (game) and 30110 (HTTP)
- Server license is valid
- Server not at max capacity
- Client FiveM is up to date

Q38: How do you fix database errors?

A:

- Verify MySQL service is running
- Check connection string in server.cfg
- Ensure user has proper permissions
- Verify tables exist (import sql.sql)
- Check for connection pool exhaustion
- Review oxmysql logs

Q39: What if a resource won't load?

A:

- Check fxmanifest.lua for syntax errors
- Verify all dependencies are started
- Ensure resource is in correct folder
- Check for duplicate resource names
- Review server console for specific errors
- Try refresh and ensure [resource]

Academic Questions

Q40: What did you learn from this project?

A:

- Lua programming and game scripting
- Client-server architecture design
- Database design and optimization
- Real-time multiplayer system development
- API design and integration
- Version control with Git
- Project management and teamwork
- Problem-solving in complex environments

Q41: How does this relate to your coursework?

A: This project applies concepts from:

- Database Systems (MySQL, schema design, optimization)
- Software Engineering (modular design, version control, testing)
- Computer Networks (client-server, protocols, latency)
- Web Development (HTML/CSS/JS for UI)
- Operating Systems (process management, resource allocation)
- Security (authentication, validation, encryption)

Q42: What makes this a good capstone project?

A:

- Large-scale system integration (50+ modules)
- Real-world application
- Multiple technical domains
- Problem-solving requirements
- Scalability considerations
- Security implementation
- Performance optimization
- Documentation and presentation

Q43: Could this be expanded for research?

A: Yes, potential research areas:

- Multiplayer synchronization algorithms
- Database optimization for gaming
- Anti-cheat detection methods
- Scalable server architectures
- Player behavior analytics
- Virtual economy modeling

Miscellaneous Questions

Q44: Do you need GTA V to play?

A: Yes, players need:

- Legal copy of GTA V (Steam/Rockstar version)
- FiveM client installed
- Server connection details

Q45: Is FiveM legal?

A: Yes, FiveM is legal and operates with Rockstar's blessing for non-commercial use. Server owners must follow Rockstar's policies.

Q46: Can console players join?

A: No, FiveM is PC-only. GTA Online is separate from FiveM.

Q47: How much does it cost to run?

A:

- FiveM license: Free
- Server hosting: \$10-50/month (depending on specs)

- Development time: [Your estimate]
- Domain/website: Optional, \$10-20/year

Q48: Where can I learn more?

A:

- FiveM Documentation: <https://docs.fivem.net/>
- Qbox Framework: <https://github.com/Qbox-project>
- Overextended: <https://overextended.dev/>
- FiveM Forums: <https://forum.cfx.re/>
- Discord communities for each framework

Q49: Can I get the source code?

A: [Specify based on your policy - e.g., "For academic purposes and with proper attribution, yes. Contact [email/contact info]."]

Q50: Who maintains this server?

A: [Your team information]

- Team Members: [Names]
- Project Supervisor: [Name]
- Institution: NSU (North South University)
- Course: CSE499
- Contact: [Email/Discord]

Contact for More Questions

If your question isn't answered here, please contact:

Email: [your-email@example.com]

Discord: [Your Discord]

GitHub: [Repository link if applicable]

Or visit our presentation materials in the Final_Presentation folder.

Document Version: 1.0

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6. Complete Feature List

Complete Feature List

NSU CSE499 - GTA V Roleplay Server

Overview

This document provides a comprehensive list of all features implemented in the NSU CSE499 GTA V Roleplay Server.

Total Resources: 50+

Total Features: 100+

Framework: Qbox (QBX)

Platform: FiveM

Core Framework Features

Player Management

- ■ Multi-character system (multiple characters per account)
- ■ Character creation with full customization
- ■ Character selection screen
- ■ Persistent player data across sessions
- ■ Player spawning system
- ■ Respawn mechanics
- ■ Character deletion

Job System

- ■ Dynamic job assignment
- ■ Multi-job support (multiple jobs per player)

- ■ Job grades and promotions
- ■ Job-specific permissions
- ■ On/off duty toggle
- ■ Boss menus for job leaders
- ■ Job-specific garages
- ■ Salary payments
- ■ 10+ unique job roles

Gang System

- ■ Gang creation and management
- ■ Gang hierarchy
- ■ Territory control
- ■ Gang-specific features
- ■ Gang stashes

Economy

- ■ Cash system
- ■ Bank accounts
- ■ ATM access
- ■ Banking UI (Renewed-Banking)
- ■ Transaction history
- ■ Account management
- ■ Job accounts
- ■ Gang accounts
- ■ Shared accounts

Inventory & Items

Inventory System (ox_inventory)

- ■ 50 item slots
- ■ 85kg weight capacity

- ■ Item metadata support
- ■ Drag and drop interface
- ■ Item stacking
- ■ Item durability
- ■ Item tooltips
- ■ Search functionality
- ■ Quick use items
- ■ Item drops (3D props)

Item Categories

- ■ Weapons (pistols, rifles, melee)
- ■ Weapon attachments (scopes, suppressors, grips)
- ■ Ammunition
- ■ Food and drinks
- ■ Medical items
- ■ Tools (lockpicks, repair kits)
- ■ Job-specific items
- ■ Vehicle keys
- ■ Phone
- ■ Radio
- ■ Crafting materials

Storage Systems

- ■ Personal stashes
- ■ Vehicle trunks
- ■ Vehicle gloveboxes
- ■ Property storage
- ■ Job storage
- ■ Gang stashes
- ■ Evidence lockers
- ■ Shops

Vehicle Systems

Vehicle Management

- ■ Vehicle ownership
- ■ Vehicle spawning
- ■ Vehicle storage (garages)
- ■ Multiple garage locations
- ■ Impound system
- ■ Vehicle state persistence
- ■ Vehicle metadata

Vehicle Features

- ■ Vehicle keys system
- ■ Lockpicking vehicles
- ■ Fuel system (ox_fuel)
- ■ Vehicle damage persistence
- ■ Engine toggle
- ■ Vehicle sharing/lending
- ■ GPS/waypoint system

Vehicle Customization

- ■ Vehicle mod shop (qbx_customs)
- ■ Performance upgrades
- ■ Visual customization
- ■ Paint jobs
- ■ Wheels and tires
- ■ Neon lights
- ■ Window tint
- ■ License plates

Vehicle Shop

- ■ Vehicle dealership

- ■ Browse vehicles by category
- ■ Test drives
- ■ Purchase system
- ■ Financing options
- ■ Vehicle sales (player-to-player)

Racing

- ■ Street racing system
- ■ Lap racing tracks
- ■ Race creation
- ■ Leaderboards
- ■ Race betting
- ■ Checkpoints system

Property System

Property Features

- ■ Property ownership
- ■ Property rentals
- ■ Multiple property types
- ■ Property storage
- ■ Property keys
- ■ Property sharing
- ■ Interior customization
- ■ Property upgrades

Property Types

- ■ Apartments
- ■ Houses
- ■ Garages
- ■ Warehouses

- ■ Businesses

Communication Systems

Phone System (NPWD)

- ■ Full smartphone UI
- ■ Home screen
- ■ App drawer
- ■ Notification system
- ■ Settings menu

Phone Apps

- ■ **Contacts** - Contact management
- ■ **Phone** - Make/receive calls
- ■ **Messages** - Text messaging
- ■ **Camera** - Take photos, selfies
- ■ **Gallery** - View saved photos
- ■ **Garage** - Vehicle management
- ■ **Mail** - In-game mail system
- ■ **Settings** - Phone customization
- ■ **Marketplace** - [If enabled]
- ■ Photo upload (Fivemanage/Imgur)

Voice Chat (pma-voice)

- ■ Proximity voice (3D positional)
- ■ Voice range adjustment (whisper/normal/shout)
- ■ Radio communication
- ■ Phone calls
- ■ Voice fade with distance
- ■ Discord integration
- ■ Multiple radio channels

- ■ Radio click sounds
- ■ Push-to-talk

Radio System

- ■ Handheld radio item
- ■ Multiple channels (1-999)
- ■ Channel encryption
- ■ Radio UI
- ■ Job-specific channels

Job Systems (Detailed)

1. Police Department (qbx_police)

- ■ Duty system
- ■ Police armory
- ■ Police garage (vehicles)
- ■ Handcuffs
- ■ Jail system
- ■ Prisoner transport
- ■ Evidence collection
- ■ DNA sampling
- ■ Fingerprints
- ■ Bullet casings
 - ■ Evidence lockers
 - ■ Radar gun (speed detection)
 - ■ Spike strips
 - ■ Police computer/MDT
 - ■ Wanted system
 - ■ Warrant system
 - ■ Mugshots
 - ■ Seizure system (items/vehicles)

- ■ Police alerts for crimes
- ■ Backup calling system
- ■ Police stations

2. EMS/Ambulance (qbx_ambulancejob)

- ■ Duty system
- ■ EMS garage (ambulances)
- ■ Medical equipment
- ■ Revive system
- ■ Heal players
- ■ Diagnose injuries
- ■ Hospital check-in
- ■ Death system
- ■ Respawn mechanics
- ■ Medical supplies
- ■ EMS alerts
- ■ Hospital beds
- ■ Medical bills

3. Mechanic (qbx_mechanicjob)

- ■ Vehicle repair
- ■ Engine repair
- ■ Body repair
- ■ Performance tuning
- ■ Vehicle cleaning
- ■ Tire changes
- ■ Mechanic shop
- ■ Mobile repairs
- ■ Tow truck integration
- ■ Parts system

4. Taxi Driver (qbx_taxijob)

- ■ Taxi vehicles

- ■ Fare system
- ■ NPC fares
- ■ Player fares
- ■ Taxi meter
- ■ GPS routing
- ■ Tips system
- ■ Taxi blips

5. Bus Driver (qbx_busjob)

- ■ Bus routes
- ■ Bus stops
- ■ Passenger system
- ■ Route payments
- ■ Bus garage

6. Truck Driver (qbx_truckerjob)

- ■ Cargo delivery
- ■ Trailer attachment
- ■ Delivery routes
- ■ Payment based on distance
- ■ Multiple truck types
- ■ Cargo variety

7. Garbage Collector (qbx_garbagejob)

- ■ Garbage truck
- ■ Trash collection points
- ■ Route system
- ■ Bag collection
- ■ Disposal system

8. Tow Truck Driver (qbx_towjob)

- ■ Tow truck vehicles
- ■ Vehicle towing

- ■ Impound lot
- ■ Tow requests
- ■ NPC vehicle towing
- ■ Payment system

9. News Reporter (qbx_newsjob)

- ■ Camera equipment
- ■ Microphone
- ■ News vehicles
- ■ Live reporting
- ■ Photo journalism

10. Lawyer

- ■ Lawyer office
- ■ Reduce jail time
- ■ Legal representation
- ■ Court integration

11. Real Estate

- ■ Property listings
- ■ Property sales
- ■ Commissions
- ■ Property tours

12. Pawn Shop (qbx_pawnshop)

- ■ Sell items
- ■ Item valuation
- ■ Cooldown system

13. Car Wash (qbx_carwash)

- ■ Vehicle cleaning
- ■ Self-service option

- ■ Employee service
- ■ Payment system

Criminal Activities

Robberies

Bank Robbery (qbx_bankrobbery)

- ■ Multiple bank locations
- ■ Thermite hacking
- ■ Vault access
- ■ Loot system
- ■ Police alerts
- ■ Cooldown system
- ■ Difficulty levels

Store Robbery (qbx_storerobbery)

- ■ 24/7 stores
- ■ LiquorStores
- ■ Register robbery
- ■ Safe cracking
- ■ Police alerts
- ■ Bag system

House Robbery (qbx_houserobbery)

- ■ Multiple houses
- ■ Lockpicking
- ■ Item stealing
- ■ Security systems
- ■ Police alerts

Truck Robbery (qbx_truckrobbery)

- ■ Armored trucks
- ■ Cargo trucks
- ■ Chase mechanics
- ■ Loot rewards

Drug System (qbx_drugs)

- ■ Weed production (qbx_weed)
- ■ Drug dealing
- ■ Drug effects
- ■ Drug territories
- ■ Police seizure
- ■ Corner selling

Other Criminal Activities

- ■ Vehicle theft
- ■ Lockpicking
- ■ Safecracking minigame
- ■ Hacking minigame (mhacking)
- ■ Fleeca jobs
- ■ Money laundering

Interaction Systems

ox_target

- ■ Point-and-click interactions
- ■ Entity targeting
- ■ Player targeting
- ■ Vehicle targeting
- ■ Object targeting
- ■ Zone targeting
- ■ Custom options per target

- ■ Distance checking
- ■ Job-specific options
- ■ Item requirement checking

Radial Menu (qbx_radialmenu)

- ■ Quick access menu
- ■ Vehicle options
- ■ General actions
- ■ Job actions
- ■ Emotes
- ■ Customizable options

Emotes (scully_emotemenu)

- ■ 100+ animations
- ■ Search functionality
- ■ Favorites system
- ■ Walk styles
- ■ Shared emotes (2-player)
- ■ Prop emotes
- ■ Categories

UI/HUD Systems

HUD (qbx_hud)

- ■ Health bar
- ■ Armor bar
- ■ Hunger indicator
- ■ Thirst indicator
- ■ Stress indicator
- ■ Stamina bar
- ■ Oxygen meter (swimming)

- ■ Voice indicator
- ■ Money display
- ■ Job display
- ■ Minimap
- ■ Compass
- ■ Street names
- ■ Vehicle HUD (speed, fuel, etc.)

Notifications (ox_lib)

- ■ Success notifications
- ■ Error notifications
- ■ Info notifications
- ■ Warning notifications
- ■ Custom duration
- ■ Positioning options

Progress Bars (ox_lib)

- ■ Customizable duration
- ■ Cancellable progress
- ■ Animation integration
- ■ Disable controls during progress
- ■ Custom labels

Dialogs & Menus (ox_lib)

- ■ Input dialogs
- ■ Context menus
- ■ Alert dialogs
- ■ Confirmation prompts
- ■ Select menus
- ■ Multi-select options

Character Customization (illenium-appearance)

- ■ Face features

- ■ Hair styles and colors
- ■ Tattoos
- ■ Makeup
- ■ Clothing
- ■ Accessories
- ■ Save/load outfits
- ■ Outfit shops

Scoreboard (qbx_scoreboard)

- ■ Player list
- ■ Player count
- ■ Server info
- ■ Ping display
- ■ Job display

Administrative Tools

Admin Menu (qbx_adminmenu)

- ■ Player management
- ■ Kick players
- ■ Ban players
- ■ Warn players
- ■ Freeze players
- ■ Spectate players
- ■ Revive players
- ■ Heal players
- ■ Set job
- ■ Give money
- ■ Give items

- ■ Teleportation
- ■ Teleport to player
- ■ Bring player
- ■ Teleport to coords
- ■ Teleport to waypoint
- ■ Vehicle management
- ■ Spawn vehicle
- ■ Delete vehicle
- ■ Fix vehicle
- ■ Refuel vehicle
- ■ Server management
- ■ Announcements
- ■ Weather control
- ■ Time control
- ■ Resource restart
- ■ Noclip mode
- ■ God mode
- ■ Invisibility
- ■ Permission levels

Permission System (permissions.cfg)

- ■ Group-based permissions
- ■ Individual permissions
- ■ ACE permission system
- ■ Principal assignments
- ■ Inheritance

Logging System

- ■ Discord webhooks
- ■ Player join/quit logs
- ■ Death logs

- ■ Resource logs
- ■ Transaction logs
- ■ Admin action logs
- ■ Chat logs
- ■ Vehicle logs

Ban System

- ■ Permanent bans
- ■ Temporary bans
- ■ Multi-identifier tracking
- ■ Reason logging
- ■ Banned by tracking
- ■ Ban appeals (configurable)

Additional Features

Weather & Time

- ■ Synchronized weather (Renewed-Weathersync)
- ■ Dynamic weather changes
- ■ Time synchronization
- ■ Admin weather control
- ■ Realistic transitions

Environment

- ■ IPL loader (custom interiors)
- ■ Custom maps
- ■ Density control (AI, traffic, parked vehicles)
- ■ Population control

Sound System (mana_audio)

- ■ Custom audio playback
- ■ 3D positional audio
- ■ Vehicle audio
- ■ Environment sounds

Binoculars (qbx_binoculars)

- ■ Binocular item
- ■ Zoom functionality
- ■ Night vision mode

ID Cards (qbx_idcard)

- ■ Driver license
- ■ ID card
- ■ Weapon license
- ■ Show to players

Small Resources (qbx_smallresources)

- ■ AFK kick
- ■ Discord rich presence
- ■ Map blips
- ■ Density adjustments
- ■ Weapon draw
- ■ Pointing
- ■ Ragdoll
- ■ Seatbelt
- ■ Speed limiter
- ■ Cruise control
- ■ Tackle
- ■ Crouch/prone

Door Lock System (ox_doorlock)

- ■ Lock/unlock doors
- ■ Job-restricted doors

- ■ Key card system
- ■ Multiple door types
- ■ Remote door control

Loadscreen

- ■ Custom loading screen
- ■ Server branding
- ■ Logo display
- ■ Loading tips
- ■ Music (optional)

Chat System

- ■ Chat messages
- ■ Commands
- ■ Staff chat
- ■ Admin chat
- ■ Job chat
- ■ /me, /do, /ooc commands
- ■ Chat suggestions
- ■ Chat formatting

Minigames

- ■ Safecracker minigame
- ■ Hacking minigame (mhacking)
- ■ Lockpick minigame
- ■ Thermite minigame

Vehicle Packs (Assets)

Police Vehicles (BD_PD_CAR_PACK)

- ■ Police cruisers
- ■ Unmarked cars
- ■ SUVs
- ■ Motorcycles
- ■ SWAT vehicles
- ■ Custom liveries
- ■ Emergency lights

EMS Vehicles (BD_EMS_CAR_PACK)

- ■ Ambulances
- ■ Fire trucks
- ■ Paramedic vehicles
- ■ Custom liveries
- ■ Emergency lights

General Vehicles

- ■ Civilian cars
- ■ Sports cars
- ■ SUVs
- ■ Trucks
- ■ Motorcycles
- ■ Boats
- ■ Aircraft

Performance & Optimization Features

Database

- ■ Connection pooling
- ■ Async queries (non-blocking)
- ■ Transaction support
- ■ Prepared statements

- ■ Indexed tables

Server

- ■ Resource streaming
- ■ Optimized scripts
- ■ Event throttling
- ■ Rate limiting
- ■ Memory management
- ■ Lua 5.4 optimization

Network

- ■ Bandwidth optimization
- ■ Entity culling
- ■ Network event validation

Security Features

Anti-Cheat

- ■ Server-side validation
- ■ Event validation
- ■ Weapon validation
- ■ Money validation
- ■ Teleport detection
- ■ Speed hack detection
- ■ God mode detection

SQL Security

- ■ Prepared statements
- ■ Parameter binding
- ■ Input sanitization

- ■ No string concatenation

Authorization

- ■ Permission checking
- ■ Job verification
- ■ Level verification
- ■ Distance checking
- ■ Rate limiting

Configuration Features

Customizable Settings

- ■ Server name
- ■ Max players
- ■ Starting money
- ■ Job salaries
- ■ Vehicle prices
- ■ Property prices
- ■ Respawn locations
- ■ Job locations
- ■ Shop locations
- ■ Robbery locations
- ■ Item metadata
- ■ Weapon damage
- ■ Hunger/thirst rates
- ■ Voice ranges

Localization

- ■ Multi-language support
- ■ Locale files for each resource
- ■ Translation system

Developer Features

API/Exports

- ■ GetPlayer export
- ■ GetPlayers export
- ■ Player functions
- ■ Money functions
- ■ Item functions
- ■ Vehicle functions
- ■ Job functions
- ■ Custom callbacks

Events

- ■ Player loaded event
- ■ Player unload event
- ■ Job update event
- ■ Money change event
- ■ Resource start/stop events
- ■ Custom event system

Hooks

- ■ Resource hooks
- ■ Player hooks
- ■ Vehicle hooks
- ■ Inventory hooks

Summary Statistics

Total Implemented Features: 100+

Framework: Qbox (QBX)

Status: Production Ready

Version: 1.0

Document Version: 1.0

Last Updated: [Date]

For: NSU CSE499 Final Presentation

7. Installation & Setup Guide

Installation & Setup Guide

NSU CSE499 - GTA V Roleplay Server

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1. Prerequisites

Required Software

- [] **MySQL Server** (8.0 or MariaDB 10.6+)
- [] **FiveM Server** (Latest artifact)
- [] **Git** (for version control)
- [] **Text Editor** (VS Code recommended)

Required Accounts

- [] **FiveM Account** - <https://keymaster.fivem.net/>
- [] **Discord Account** (for webhooks)
- [] **Rockstar Games Account** (for testing)

Knowledge Prerequisites

- Basic command line usage
- Basic SQL knowledge
- Text file editing
- Port forwarding (if hosting at home)

2. System Requirements

Minimum Requirements

- **CPU:** Intel Core i5-4590 / AMD FX-8350 (4 cores @ 3.0 GHz)
- **RAM:** 8 GB
- **Storage:** 50 GB SSD
- **Network:** 100 Mbps down/up
- **OS:** Windows Server 2019 / Ubuntu 20.04

Recommended Requirements

- **CPU:** Intel Core i7-8700 / AMD Ryzen 7 2700X (8 cores @ 3.5 GHz)
- **RAM:** 16 GB
- **Storage:** 100 GB NVMe SSD
- **Network:** 1 Gbps down/up
- **OS:** Windows Server 2022 / Ubuntu 22.04 LTS

Network Requirements

- **Ports to Open:**
 - 30120 (TCP/UDP) - Game server
 - 30110 (TCP) - HTTP server
 - 3306 (TCP) - MySQL (if remote)
 - 40120 (TCP) - txAdmin (optional)

3. Installation Steps

Step 1: Install MySQL/MariaDB

Windows:

1. Download MySQL from <https://dev.mysql.com/downloads/installer/>
2. Run installer
3. Choose "Server only" installation
4. Set root password (remember this!)
5. Complete installation

Linux (Ubuntu):

```
# Update package list
sudo apt update

# Install MySQL Server
sudo apt install mysql-server -y

# Secure installation
sudo mysql_secure_installation
```

Step 2: Create Database

```
# Login to MySQL
mysql -u root -p

# In MySQL console:
CREATE DATABASE qbox_27adff CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
CREATE USER 'qbox_user'@'localhost' IDENTIFIED BY 'YOUR_STRONG_PASSWORD';
GRANT ALL PRIVILEGES ON qbox_27adff.* TO 'qbox_user'@'localhost';
FLUSH PRIVILEGES;
EXIT;
```

Step 3: Import Database Schema

```
# Navigate to project directory
cd E:\FiveM_Server\NSU_CSE499

# Import SQL file
mysql -u qbox_user -p qbox_27adff < sql.sql
```

```
# Enter password when prompted
```

Verify Import:

```
mysql -u qbox_user -p qbox_27adff  
SHOW TABLES;  
# Should show: bank_accounts, bans, dealers, lapraces, etc.  
EXIT;
```

Step 4: Download FiveM Server

Windows:

1. Go to https://runtime.fivem.net/artifacts/fivem/build_server_windows/master/
2. Download latest server.zip
3. Extract to C:\FXServer\

Linux:

```
# Create directory  
mkdir -p /home/fivem/server  
cd /home/fivem/server  
  
# Download latest artifact  
wget https://runtime.fivem.net/artifacts/fivem/build_proot_linux/master/[BUILD_NUMBER]  
/fx.tar.xz  
  
# Extract  
tar xf fx.tar.xz  
  
# Make executable  
chmod +x run.sh
```

Step 5: Get FiveM License Key

1. Visit <https://keymaster.fivem.net/>
2. Login with your FiveM account
3. Click "Generate License Key"
4. Fill in server details:
 - **Server Name:** NSU CSE499 Development
 - **Server IP:** Leave blank (optional)

- **Type:** Server

5. Copy the generated key (starts with cfxk_)

Step 6: Configure Server Files

server.cfg

Open server .cfg and update:

```
# License Key (REQUIRED)
sv_licenseKey "YOUR_LICENSE_KEY_HERE"

# MySQL Connection (REQUIRED)
set mysql_connection_string
"mysql://qbox_user:YOUR_PASSWORD@localhost/qbox_27adff?connectionLimit=10"

# Server Settings
sv_hostname "NSU CSE499 Server"
sv_maxclients 10
set sv_projectName "NSU CSE499"
set sv_projectDesc "FiveM Roleplay Server"

# Server Admins (Add your identifiers)
add_ace group.admin command allow
add_ace group.admin command.quit deny
add_principal identifier.license:YOUR_LICENSE_HERE group.admin
```

Find Your License Identifier:

1. Connect to your server
2. Check server console
3. Look for line: [system] Player connecting: [Name] (license:XXXXXXXXX)
4. Copy the license identifier

Step 7: Verify Resource Structure

Ensure folder structure looks like:

```
E:\FiveM_Server\NSU_CSE499\
■■■■ resources/
■ ■■■■ [qbx]/
■ ■■■■ [ox]/
■ ■■■■ [standalone]/
■ ■■■■ [voice]/
■ ■■■■ [MEV]/
■ ■■■■ [cfx-default]/
■■■■ server.cfg
■■■■ ox.cfg
■■■■ voice.cfg
■■■■ misc.cfg
```

```
■■■ permissions.cfg
■■■ sql.sql
```

Step 8: Install Dependencies

All dependencies should already be in the resources folder. Verify these exist:

- resources/[ox]/ox_lib/
- resources/[ox]/oxmysql/
- resources/[ox]/ox_inventory/
- resources/[qb]/qb_core/

4. Configuration

Discord Webhooks (Optional but Recommended)

1. **Create Discord Server** (if you don't have one)
2. **Create Webhook:**
 - Go to Server Settings → Integrations → Webhooks
 - Click "New Webhook"
 - Name it (e.g., "Server Logs")
 - Copy webhook URL

3. **Add to Configuration:**

Edit relevant config files (e.g., resources/[qb]/qb_core/config/server.lua):

```
Config.DiscordWebhook = "YOUR_WEBHOOK_URL_HERE"
```

Admin Permissions

Edit permissions.cfg:

```
# Add yourself as admin
add_principal identifier.license:YOUR_LICENSE_HERE group.admin

# Add team members
add_principal identifier.license:TEAMMATE_LICENSE_1 group.admin
add_principal identifier.license:TEAMMATE_LICENSE_2 group.mod
```

Starting Money

Edit resources/[qbx]/qbx_core/config/shared.lua:

```
Config.Money = Config.Money or {}
Config.Money.MoneyTypes = {
    cash = 5000, -- Starting cash
    bank = 5000, -- Starting bank balance
    crypto = 0 -- Starting crypto
}
```

Job Salaries

Edit resources/[qbx]/qbx_core/shared/jobs.lua:

```
['police'] = {
    grades = {
        [0] = { name = 'Cadet', payment = 50 },
        [1] = { name = 'Officer', payment = 75 },
        [2] = { name = 'Sergeant', payment = 100 },
        -- etc.
    }
}
```

5. First Launch

Windows

1. Navigate to FiveM Server folder:

```
cd C:\FXServer
```

2. Copy your server.cfg to server folder:

```
copy "E:\FiveM_Server\NSU_CSE499\server.cfg" "C:\FXServer\server.cfg"
```

3. Start Server:

```
FXServer.exe +exec server.cfg
```

Linux

```
cd /home/fivem/server  
  
# Start server  
./run.sh +exec server.cfg
```

What to Look For

Successful Startup:

```
/ _ \ (-) / _ \| V |  
| | | | | | | | | \ / | | |
| | | | | | | | | V | |  
| | | | | | | | | | | |  
\_ \ \ \ \\_ \| | | | |  
  
citizenfx.net
```

Server started on port 30120
Resources loaded
MySQL connection successful
qbx_core started
All resources ready

```
/ _ \ (-) / _ \| V |  
| | | | | | | | | \| / |  
| | | | | | | | | \| V |  
| | | | | | | | | \| V |  
\ _ \| \_ \| \_ \| | | | |  
citizenfx.net
```

Server started on port 30120
Resources loaded
MySQL connection successful
qbx_core started
All resources ready

Common Errors:

- Invalid license key → Check `sv_licenseKey`
- MySQL connection failed → Check `mysql_connection_string`
- Failed to load resource → Check resource dependencies

6. Testing

Connect as Player

1. **Open FiveM Client**
2. **Press F8** (console)
3. **Type:**

```
connect localhost
```

Or if remote:

```
connect YOUR_SERVER_IP:30120
```

Test Checklist

- ☐ Server loads without errors
- ☐ Can connect to server
- ☐ Character creation works
- ☐ Can spawn in game
- ☐ Inventory opens (Tab/I)
- ☐ Phone opens (P/M)
- ☐ Voice chat works (test with second player)
- ☐ Job system works (/job)
- ☐ Vehicle spawn works (if admin: /car adder)
- ☐ Database saves (logout and login again)

Admin Commands Test

```
/admin - Open admin menu  
/noclip - Toggle noclip  
/god - Toggle god mode  
/car adder - Spawn vehicle  
/setjob [id] police 2 - Set job
```

Multi-Player Test

1. Connect with 2+ clients
2. Test voice proximity
3. Test phone calls
4. Test vehicle interactions
5. Test job features (police arrest, etc.)

7. Troubleshooting

Server Won't Start

Issue: Server crashes immediately

Solution:

1. Check server console for errors
2. Verify license key is correct
3. Ensure all config files have proper syntax
4. Check port 30120 isn't already in use

Database Connection Failed

Issue: MySQL connection error

Solution:

1. Verify MySQL is running:
Windows: Check Services
Linux: `sudo systemctl status mysql`
2. Test connection manually:
`mysql -u qbox_user -p qbox_27adff`
3. Check connection string format:
`mysql://user:password@host/database`

Can't Connect to Server

Issue: "Server not found"

Solution:

1. Verify server is running
2. Check firewall allows port 30120
3. Try connecting with IP:PORT explicitly
4. Check logs for "Server started on port 30120"

Resources Not Loading

Issue: Resources show errors

Solution:

1. Check resource dependencies in fxmanifest.lua
2. Ensure dependencies load first (check server.cfg order)
3. Try: refresh, ensure [resource]
4. Check resource folder structure

Voice Not Working

Issue: Can't hear other players

Solution:

1. Verify pma-voice is started
2. Check voice.cfg is executed
3. Test voice key (default: N)
4. Check Discord settings if using Discord integration
5. Ensure voice resource starts before others

Character Not Saving

Issue: Progress lost on disconnect

Solution:

1. Check database connection
2. Verify qbx_core is running
3. Check for errors in server console
4. Test direct SQL query to verify connection

8. Maintenance

Regular Backups

Database Backup (Daily Recommended)

Windows:

```
@echo off
set TIMESTAMP=%date:~-4,4%%date:~-10,2%%date:~-7,2%_%time:~0,2%%time:~3,2%%time:~6,2%
set TIMESTAMP=%TIMESTAMP: =0%
mysqldump -u qbox_user -p qbox_27adff > backup_%TIMESTAMP%.sql
```

Linux (Cron Job):

```
# Create backup script
nano /home/fivem/backup.sh

# Add:
#!/bin/bash
mysqldump -u qbox_user -pYOUR_PASSWORD qbox_27adff > /home/fivem/backups/backup_$(date +%Y%m%d_%H%M%S).sql

# Make executable
chmod +x /home/fivem/backup.sh

# Add to crontab (daily at 3 AM)
crontab -e
0 3 * * * /home/fivem/backup.sh
```

Resource Backup

```
# Compress resources folder
tar -czf resources_backup_$(date +%Y%m%d).tar.gz resources/
```

Updates

Update Server Artifacts

```
# Download latest build
# Extract to server folder
# Restart server
```

Update Resources

```
# Backup current version
# Replace resource folder
# Check for config changes
# Test on development server first
# Deploy to production
```

Monitoring

Check Server Status

```
# Windows (PowerShell)
Get-Process FXServer

# Linux
ps aux | grep FXServer
```

Monitor Logs

```
# Server console shows real-time logs
# Check for errors regularly
# Set up Discord webhooks for critical errors
```


Performance Optimization

Database Optimization (Monthly)

```
# Optimize tables
OPTIMIZE TABLE characters;
OPTIMIZE TABLE player_vehicles;
OPTIMIZE TABLE bank_accounts;

# Analyze tables
ANALYZE TABLE characters;
```

Clear Old Data (Optional)

```
# Delete bans older than 6 months
DELETE FROM bans WHERE expire < UNIX_TIMESTAMP(DATE_SUB(NOW(), INTERVAL 6 MONTH)) AND
expire > 0;

# Clear old logs (if logging table exists)
```

9. Production Deployment Checklist

Before Going Live

• ☐ Security

- ☐ Strong MySQL password
- ☐ Firewall configured
- ☐ Admin permissions set
- ☐ Discord webhooks configured
- ☐ Rate limiting enabled

• ☐ Performance

- ☐ Database optimized and indexed
- ☐ Connection pooling configured
- ☐ Resource limits set
- ☐ Backup system active

- ☐ **Configuration**

- ☐ Server name set
- ☐ Max players configured
- ☐ Starting money set
- ☐ Job salaries balanced
- ☐ Spawn points tested

- ☐ **Testing**

- ☐ All jobs tested
- ☐ Economy tested
- ☐ Voice chat tested
- ☐ Phone system tested
- ☐ Admin tools tested
- ☐ Multi-player tested

- ☐ **Documentation**

- ☐ Admin guide written
- ☐ Player rules created
- ☐ Command list available
- ☐ Support system ready

- ☐ **Monitoring**

- ☐ Logging enabled
- ☐ Performance monitoring
- ☐ Error tracking
- ☐ Player feedback system

10. Quick Reference

Essential Commands

Server Console:

```
status # Show players
restart [resource] # Restart resource
refresh # Refresh resource list
quit # Stop server
```

In-Game (Admin):

```
/admin # Admin menu
/noclip # Noclip mode
/car [model] # Spawn vehicle
/dv # Delete vehicle
/setjob [id] [job] [grade] # Set job
/givemoney [id] [type] [amount] # Give money
/revive # Revive self
/revive [id] # Revive player
```

File Locations

```
Server Config: server.cfg
Database Schema: sql.sql
Core Config: resources/[qbx]/qbx_core/config/
Inventory Config: resources/[ox]/ox_inventory/data/
Admin Perms: permissions.cfg
```

Support Resources

- **FiveM Docs:** <https://docs.fivem.net/>
- **Qbox GitHub:** <https://github.com/Qbox-project>
- **Overextended:** <https://overextended.dev/>
- **FiveM Forums:** <https://forum.cfx.re/>

Congratulations!

Your NSU CSE499 GTA V Roleplay Server should now be running!

For questions or issues:

- Check troubleshooting section

- Review logs in server console
- Consult official documentation

Document Version: 1.0

Last Updated: [Date]

For: NSU CSE499 Final Presentation